Machine Learning for Semiconductor Yield Prediction: A Comparative Study with SHAP Interpretability Analysis on the SECOM Dataset

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***Abstract*—Semiconductor yield prediction from high-dimensional process data is challenged by severe class imbalance, widespread missing sensor readings, and failure signals distributed across hundreds of process variables rather than concentrated in a few dominant signals. While machine learning methods have been applied to this problem, few studies report cross-validated performance under class imbalance or provide sensor-level explanations that process engineers can act upon. This paper presents a comparative analysis of three machine learning models — logistic regression, Random Forest, and XGBoost — for yield prediction on the SECOM semiconductor manufacturing dataset, incorporating SHAP interpretability analysis, a controlled experiment on missingness as a predictive signal, and recall-oriented threshold selection motivated by asymmetric misclassification costs. Cross-validated PR-AUC results show that performance differences between models are within noise bounds given the dataset's 104 failure samples, but XGBoost achieves substantially higher operational recall — recovering 19 of 21 test failures, including a failure masked by competing sensor signals that Random Forest missed — at the cost of a higher false-alarm rate. SHAP analysis shows that 237 of 446 features are required to account for 80% of total feature importance, indicating that predictive signal is broadly distributed rather than concentrated in a small subset. This challenges simple feature selection approaches and suggests that yield failure reflects multi-causal process interactions.**

***Index Terms*—semiconductor yield prediction, machine learning, SHAP interpretability, class imbalance, Random Forest, XGBoost, SECOM dataset, precision-recall optimization**

I. Introduction

Semiconductor manufacturing consists of hundreds of sequential process steps, each introducing variation that accumulates across the wafer. Yield — the fraction of dies passing final electrical test — directly determines fab profitability, as each defective wafer that advances through the full process sequence consumes significant time, labor, and materials before failure is detected at end-of-line test. Early identification of at-risk wafers enables intervention before these costs accumulate, motivating the application of machine learning to yield prediction from in-line process sensor data [1], [2].

The SECOM dataset presents the core challenges of real-world yield prediction: 1,567 wafer observations across 590 process measurement features with significant sensor dropout, and a 14:1 class imbalance in which only 6.64% of wafers fail [3]. A naive classifier predicting pass for all wafers achieves 93.36% accuracy while recovering zero failures — exposing the inadequacy of accuracy as an evaluation metric [4]. Failure patterns are distributed across process parameters rather than concentrated in dominant signals, meaning that standard feature selection approaches and single-metric evaluations obscure the true difficulty of this prediction task [5].

Few studies report cross-validated PR-AUC for SECOM — most report single-split accuracy or ROC-AUC, which are inadequate for this imbalance level and subject to high variance given the small failure count [6]. SHAP interpretability has been applied to semiconductor yield prediction generally [7], but rarely to explain specific missed failures or identify masking effects where competing sensor signals suppress failure prediction. To our knowledge, the hypothesis that missingness patterns carry failure signal — plausible given that sensor dropout may correlate with process anomalies — has not been systematically tested on the SECOM dataset.

The primary contributions of this paper are:

1) A cross-validated comparative study of three model classes for SECOM yield prediction, providing honest uncertainty quantification that reveals single-split results overestimate Random Forest performance by 0.079 PR-AUC.
2) SHAP-based identification of a sensor masking case in which Feature 31 suppresses failure signals from Features 59 and 103, together with evidence that XGBoost recovers this failure where Random Forest does not, consistent with gradient boosting's sequential error correction [8].
3) A controlled experiment testing whether missingness patterns carry failure signal beyond imputed values, finding no meaningful improvement (+0.0062 PR-AUC), supporting the median imputation preprocessing decision and suggesting limited value in missingness features on this dataset.

The remainder of this paper is organized as follows. Section II surveys related work on machine learning for semiconductor yield prediction. Section III describes the SECOM dataset and preprocessing decisions. Section IV presents the experimental methodology. Section V reports results. Sec-

tion VI discusses implications for semiconductor process engineering. Section VII concludes.

II. Related Work

Machine learning has been applied to semiconductor yield prediction since the early 2000s [1], with methods ranging from k-nearest neighbor fault detection [9] to deep learning approaches for variable-length sensor data [10]. Cascading classification systems [11] and ensemble methods [12] have demonstrated improvements over single-model baselines, though most studies report single-split accuracy or ROC-AUC metrics that are inadequate under severe class imbalance.

Interpretability methods have gained traction in semiconductor manufacturing. Wang and Chen [7] applied XGBoost with SHAP for yield prediction and failure analysis, while Presciuttini et al. [6] demonstrated LIME-based interpretability on the SECOM dataset. However, these studies focus on global feature importance rather than explaining specific missed failures or identifying sensor masking effects at the individual wafer level.

Class imbalance handling in manufacturing ML has been addressed through oversampling techniques such as SMOTE [13] and through native class weighting in ensemble methods [14]. Recent surveys [2], [15] identify class imbalance and missing data as the primary open challenges in semiconductor yield prediction. The present work addresses both challenges simultaneously while contributing per-wafer failure mode analysis and a controlled experiment on missingness as a predictive signal.

III. Dataset and Preprocessing

A. Dataset Description

The SECOM dataset contains 1,567 wafer observations from a semiconductor manufacturing line, each described by 590 anonymized process measurement features and a binary outcome label indicating pass or fail at final electrical test [3]. Of 1,567 wafers, 104 (6.64%) failed — establishing a severe class imbalance ratio of approximately 14:1. Additional challenges include 538 of 590 features containing missing values at an overall rate of 4.54%, with four features exceeding 90% missingness, and 116 features exhibiting zero variance across all observations (Table I). Feature values represent sensor readings from anonymized process steps; no universal threshold exists for acceptable missingness or minimum variance, requiring deliberate, documented preprocessing decisions.

TABLE I
SECOM Dataset Statistics

Property	Value
Total wafer observations	1,567
Process measurement features	590
Failed wafers	104 (6.64%)
Passed wafers	1,463 (93.36%)
Class imbalance ratio	14:1
Features with missing values	538 (91.2%)
Overall missing rate	4.54%
Zero-variance features	116 (19.7%)

B. Preprocessing Pipeline

Preprocessing followed five sequential steps, each representing a deliberate research decision with documented justification.

Step 1: Zero-variance removal. Features with identical values across all 1,567 wafers carry no discriminative signal. Exactly 116 such features were identified and removed, reducing the feature space from 590 to 474. Near-zero variance features were retained.

Step 2: High-missingness removal. Features missing in more than 50% of samples cannot be reliably imputed without introducing more noise than signal [16]. Twenty-eight features exceeded this threshold and were removed, yielding 446 features. Common thresholds in the literature range from 30% to 70%; we selected 50% as a moderate choice and examine sensitivity to it in Section V-D.

Step 3: Median imputation. Remaining features exhibited a mean missing rate of 1.6% (maximum 45.6%). Median imputation was preferred over mean imputation due to the presence of outliers from equipment anomalies [17]. KNN imputation and MICE were rejected due to computational cost and unverifiable distributional assumptions respectively. We assume Missing At Random (MAR) and document this as a limitation. Feature filtering and imputation were applied to the full dataset before splitting; because these steps do not use outcome labels, leakage risk is small, although fitting them within each fold would be stricter.

Step 4: Stratified train/test split. An 80/20 stratified split preserved the 6.64% failure rate in both sets, yielding 1,253 training samples (83 failures) and 314 test samples (21 failures). Without stratification, random partitioning risks substantially unequal failure rates between training and test sets.

Step 5: Feature scaling. StandardScaler was applied within the cross-validation loop — fitted on training folds only and applied to test folds — to prevent data leakage from test set statistics into scaling parameters [18].

C. Evaluation Framework

Model performance is evaluated using Precision-Recall Area Under the Curve (PR-AUC) as the primary metric [4]. Accuracy is explicitly rejected: a classifier predicting pass for all wafers achieves 93.36% accuracy while recovering zero failures — a result that appears strong but is operationally worthless in a semiconductor fab where each missed failure incurs the full cost of remaining process steps. Formally, Precision and Recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{1}$$

where TP, FP, and FN denote true positives, false positives, and false negatives respectively. PR-AUC is computed as:

$$\text{PR-AUC} = \int_0^1 P(R)\, dR \tag{2}$$

where $P(R)$ denotes precision as a function of recall across all classification thresholds.

PR-AUC is preferred over ROC-AUC for this imbalanced setting. ROC-AUC incorporates False Positive Rate, which uses the large negative class (1,463 passes) as its denominator — even substantial misclassification of passing wafers produces a small FPR, making models appear more capable than they are on the minority failure class [4]. Precision and recall are both defined with respect to the failure class — recall over actual failures and precision over predicted failures — so PR-AUC directly measures how well the model identifies the 104 failures rather than being dominated by the majority class.

Threshold selection is examined explicitly because the default classification threshold of 0.5 is inappropriate for severe class imbalance. In semiconductor manufacturing, false negatives — wafers predicted as passing that subsequently fail — consume all remaining process resources before end-of-line detection, representing the highest operational cost. False positives — correctly manufactured wafers flagged for re-inspection — incur only re-inspection cost. This asymmetric cost structure motivates lowering the classification threshold to prioritize recall, accepting additional false alarms in exchange for recovering more true failures [11]. The optimal threshold depends on fab-specific cost ratios not available for the SECOM dataset; we therefore report threshold-independent PR-AUC alongside per-wafer results at each model's selected operating threshold (Section V-B).

IV. Experimental Methodology

A. Models

Logistic regression serves as the linear baseline. Its coefficients directly indicate feature influence direction and magnitude, providing interpretability absent in ensemble methods. Class imbalance is addressed via `class_weight='balanced'`, which weights failure samples approximately 14× relative to passes during training, preventing the model from optimizing exclusively for the majority class. L2 regularization with $C = 1.0$ controls overfitting in the high-dimensional feature space (446 features, 1,253 training samples). All experiments use `random_state=42` for reproducibility.

Random Forest is selected as the nonlinear ensemble baseline [19], motivated by the hypothesis that yield failures are driven by multivariate sensor interactions that violate the linearity assumption of logistic regression. The `max_features='sqrt'` setting forces each tree to consider only $\sqrt{446} \approx 21$ randomly selected features at each split, decorrelating trees so that different trees find different patterns and make different errors — when 300 trees are averaged, these errors partially cancel. Three hundred estimators provide sufficient variance reduction for this dataset size. Trees are grown to full depth (`max_depth=None`) to assess maximum model capacity, with overfitting examined explicitly via train versus test PR-AUC comparison. Class imbalance is handled identically to logistic regression via `class_weight='balanced'`.

XGBoost is selected as the regularized gradient boosting model [20], motivated by its sequential error correction mechanism: each tree is fit to the negative gradient of the loss with respect to current ensemble predictions, so samples with large residuals receive proportionally more correction in subsequent trees. This addresses wafer-level prediction errors rather than feature-level diversity as in Random Forest. Key regularization parameters include `max_depth=4` (shallow trees preventing memorization), `learning_rate=0.05` (conservative per-tree corrections), `subsample=0.8` and `colsample_bytree=0.8` (stochastic sampling decorrelating trees), and L1/L2 regularization (`reg_alpha=0.1`, `reg_lambda=1.0`) penalizing leaf weights to limit model complexity in the high-dimensional space. Class imbalance is handled via `scale_pos_weight=14.10`, the training-set ratio of passes to failures (1,170/83), which scales the loss contribution of failure samples — amplifying the error signal so subsequent trees prioritize minority class correction. Early stopping with PR-AUC as the stopping criterion was attempted but found unreliable: with only 12 failure samples in the validation fold, PR-AUC estimation was too noisy, causing premature termination at 13 trees. A fixed 200-tree configuration trained on the full training set was therefore adopted.

B. Experimental Design

Performance is evaluated using stratified 5-fold cross-validation as the primary framework, with mean ± standard deviation of PR-AUC reported across folds [21]. A single 80/20 train/test split exhibits high variance with only 104 total failures; cross-validation averages across multiple splits to provide more reliable estimates. All models are wrapped in scikit-learn Pipelines [18] ensuring the StandardScaler is fitted only on training folds and applied to test folds, preventing data leakage. Each test fold contains approximately 21 failures, consistent with the held-out test set size.

A single held-out test set (314 samples, 21 failures) is reserved for per-wafer failure analysis and SHAP computation. Per-wafer comparison identifies which specific failure modes each model handles — a failure recovered by XGBoost but missed by Random Forest suggests that sequential error correction addressed that failure mode; failures missed by both models represent genuinely hard cases potentially requiring additional sensor data or alternative approaches [12].

SHAP (SHapley Additive exPlanations) via TreeExplainer is applied to the Random Forest to provide exact Shapley values for each wafer-feature pair [8], [22]. For wafer i and feature j, the SHAP value ϕ_{ij} is formally defined as:

$$\phi_{ij} = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{j\}) - f(S)] \quad (3)$$

where F is the full feature set, S is a feature subset, $f(S)$ is the model prediction using only features in S, and the sum is taken over all possible feature subsets excluding feature j. Positive values indicate the feature pushed failure probability upward; negative values indicate the opposite.

Global importance is computed as mean $|\phi_{ij}|$ across all test wafers.

A controlled experiment tests whether binary missingness indicators — one per feature containing any missing values in the raw data, encoding sensor dropout as a binary signal — improve yield prediction beyond imputed feature values alone [14]. XGBoost is trained on two datasets under identical 5-fold CV: preprocessed features only (Dataset A) and preprocessed features augmented with 538 missingness indicators (Dataset B). An improvement exceeding 0.01 PR-AUC is taken as evidence supporting the hypothesis that sensor dropout carries failure signal.

All code, preprocessed data, and model artifacts are publicly available at https://github.com/sudipkharel779-netizen/semiconductor-manufacturing-research.

V. Results

A. Cross-Validated Model Comparison

Table II presents 5-fold cross-validated performance across all models. XGBoost achieves the highest mean PR-AUC (0.2083 ± 0.0260), marginally outperforming Random Forest (0.1956 ± 0.0266) and Logistic Regression (0.1773 ± 0.0446). All three models achieve PR-AUC meaningfully above the naive baseline of 0.0664, demonstrating that process measurement data contains learnable failure signal despite severe class imbalance [15].

TABLE II
5-Fold Cross-Validated Performance — SECOM Dataset

Model	PR-AUC (mean±std)	ROC-AUC (mean±std)	Train PR-AUC (mean)
Naive Baseline	0.0664±0.0000	0.5000±0.0000	—
Logistic Regression	0.1773±0.0446	0.6802±0.0722	0.9629
Random Forest	0.1956±0.0266	0.7623±0.0303	1.0000
XGBoost (200 trees)	0.2083±0.0260	0.7350±0.0427	1.0000

The fold-level standard deviations of Random Forest and XGBoost overlap substantially, precluding strong claims of statistical superiority between these models at this sample size. A Wilcoxon signed-rank test on the five paired fold scores would be the appropriate significance test; however, with only five pairs the smallest attainable two-sided p-value is 0.0625, so no difference can reach conventional significance. The high standard deviation of Logistic Regression (± 0.0446) relative to ensemble methods reflects its sensitivity to which specific failures appear in each test fold — a consequence of the linear model's limited capacity to generalize across heterogeneous failure modes [23].

Both ensemble models exhibit severe train-test PR-AUC gaps under single-split evaluation (Random Forest: 0.7257, XGBoost: 0.7506), with training PR-AUC approaching 1.0 for both models. This indicates memorization of the 83 training failures rather than learning generalizable failure patterns — an expected consequence of the small failure count relative to the 446-dimensional feature space. Single-split evaluation also overstates performance: Random Forest's single-split test PR-AUC (0.2743) exceeds its cross-validated mean by 0.079.

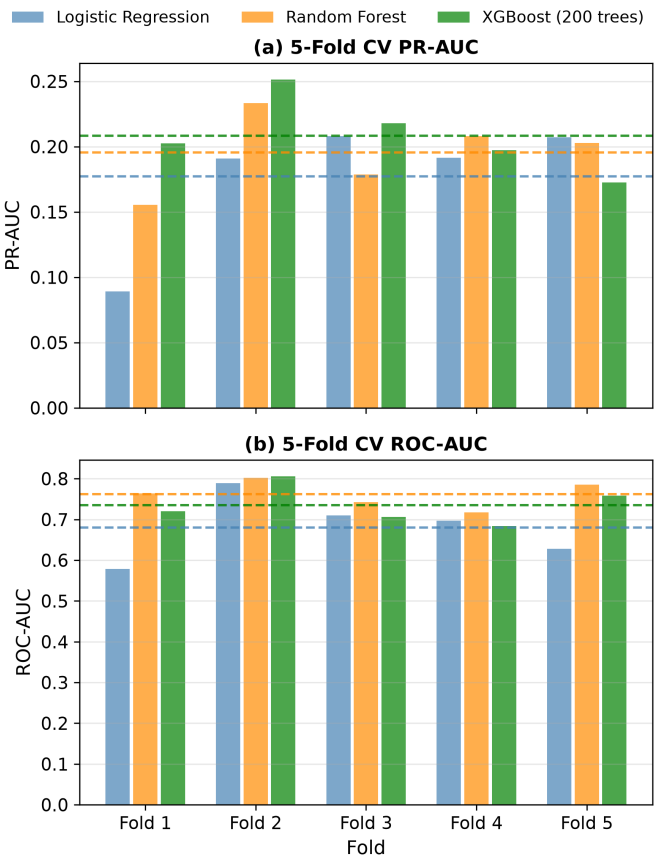


Fig. 1. 5-fold cross-validation (a) PR-AUC and (b) ROC-AUC by model. Dashed horizontal lines indicate per-model means. XGBoost achieves the highest mean PR-AUC (0.208); Random Forest achieves the highest mean ROC-AUC (0.762). Fold 1 is the hardest fold for Logistic Regression and Random Forest, while XGBoost remains near its mean, consistent with its stronger recovery of difficult failure cases.

Cross-validation confirms that the train-test gap is not specific to one split. Fig. 1 shows PR-AUC and ROC-AUC for each model across the five cross-validation folds.

B. Operational Analysis: Per-Wafer Failure Recovery

Aggregate PR-AUC conceals operationally important differences between models. In semiconductor manufacturing, the relevant question is not the average precision-recall tradeoff but how many of the 21 failed wafers in the test set were identified before consuming further processing resources. Table III presents this per-wafer analysis at each model's optimal F1 threshold.

TABLE III
Per-Wafer Failure Recovery — Single Test Set (21 Failures)

Model	Threshold	Failures Caught	False Alarms	Recall
Naive Baseline	—	0/21	0	0.000
Logistic Regression	0.55	4/21	32	0.190
Random Forest	0.20	13/21	52	0.619
XGBoost (200 trees)	0.35	19/21	183	0.905



Fig. 2. SHAP waterfall plot for Wafer 37 (true failure, missed by Random Forest, caught by XGBoost). Feature 31 ($\phi = -0.05$) generates a dominant pass-direction signal despite being elevated at 1.947 scaled units, overwhelming failure signals from Features 59 ($\phi = +0.02$) and 103 ($\phi = +0.01$).

Among 21 test failures, XGBoost recovers 19 at threshold 0.35 compared to Random Forest's 13 at threshold 0.20. Critically, XGBoost recovers all 13 failures that Random Forest catches plus 6 additional failures — Random Forest catches zero failures that XGBoost misses. This strict dominance at the individual wafer level occurs despite XGBoost's marginally lower single-split PR-AUC (0.2494 vs. 0.2743), illustrating that PR-AUC and operational recall can diverge when threshold optimization is applied to imbalanced problems [24].

Per-wafer SHAP analysis of the Random Forest identifies one of the six XGBoost-only recoveries as a masking case. In this case (test set wafer index 37), Feature 31 generates a strong pass-direction SHAP contribution ($\phi = -0.05$) despite being elevated at 1.947 scaled units, while Features 59 and 103 generate weaker failure-direction contributions ($\phi = +0.02$ and $+0.01$ respectively). In Random Forest, trees that split on Feature 31 encounter the same suppressing signal, and averaging across 300 independently trained trees dilutes the weaker failure signals, so the averaged prediction falls below the detection threshold. XGBoost's sequential boosting mechanism fits each new tree to the negative gradient of the loss function with respect to current predictions — this gradient is large for samples where the current ensemble predicts poorly, so subsequent trees implicitly allocate more correction capacity to hard-to-predict samples, a plausible explanation for its recovery of this failure [20].

Fig. 2 illustrates the SHAP waterfall for Wafer 37, showing Feature 31's dominant pass-direction contribution overcoming failure signals from Features 59 and 103. XGBoost's 19-failure recall comes at a cost of 183 false alarms (202 of 314 test wafers flagged, 64%) versus Random Forest's 52 (65 flagged, 21%). The operationally optimal model depends on the fab's cost ratio between missed failures and false alarms. If the cost of processing a failed wafer through remaining steps substantially exceeds re-inspection cost — a reasonable assumption for high-value advanced node manufacturing — the XGBoost deployment at threshold 0.35 may be economically justified [25]. Fab-specific cost parameters required to quantify this tradeoff are not available in the public SECOM dataset.

C. SHAP Interpretability Analysis

SHAP mean absolute values computed across the 314-wafer test set reveal that predictive signal is broadly distributed across the 446-feature space [22]. Exactly 237 features are required to account for 80% of total SHAP importance — far from the Pareto 80/20 concentration expected in datasets where a small number of dominant variables drive outcomes. This finding is consistent with the hypothesis that semiconductor yield failure results from genuinely multivariate process interactions rather than single-variable deviations, and has direct implications for feature selection: aggressive dimensionality reduction retaining only top-10 or top-20 features would discard the majority of predictive signal [21].

Fig. 3 shows the beeswarm distribution of SHAP values across the top 20 features. Comparison between SHAP mean absolute importance and Random Forest Gini importance reveals complete overlap in the top-10 feature rankings — all 10 features identified by SHAP appear in the Gini top-10, though ordering differs slightly. This concordance between two theoretically distinct importance measures increases confidence that the identified features carry genuine yield prediction signal rather than reflecting artifacts of either method's specific assumptions [8]. SHAP is preferred for reporting because it provides directional information and per-wafer attribution that Gini importance cannot.

The beeswarm also reveals directional effects for the top features. Feature 103 shows clear separation: high values push failure probability upward, while low values push it toward pass. Feature 59 shows a similar but more gradual relationship, with high values pushing toward failure. Feature 33 exhibits an asymmetric pattern: low values push strongly toward pass prediction, while high values show a weak mixed effect, suggesting Feature 33 operates primarily as a pass indicator rather than a failure driver. Feature 31 shows the reverse of Feature 103: high values push strongly toward pass, consistent with its suppressing role in the Wafer 37 masking case. These directional relationships are actionable for process engineers: elevated Feature 103 readings represent a candidate statistical process control signal where intervention before end-of-line test may prevent downstream yield loss [7].

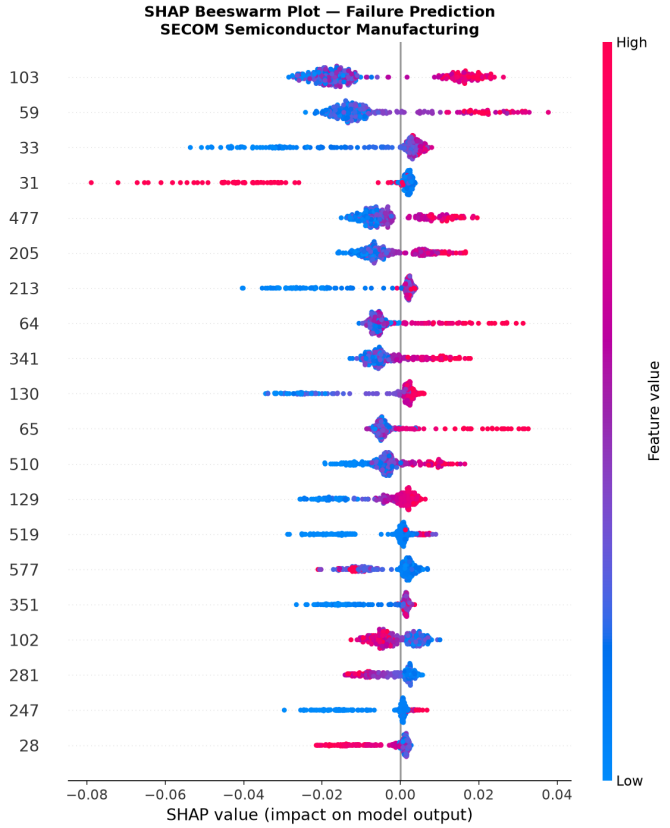


Fig. 3. SHAP beeswarm plot for top 20 features. Each dot represents one test wafer. Color indicates feature value (red = high, blue = low). Positive SHAP values push toward failure prediction. Feature 103 shows clear separation between high and low values; Feature 33 acts primarily as a pass indicator.

D. Sensitivity Analysis

To address the potential concern that reported findings reflect specific preprocessing choices rather than genuine data properties, we repeated the full preprocessing and XGBoost CV pipeline under three missingness threshold conditions: 30%, 50% (baseline), and 70%. Feature counts retained were 442, 446, and 466 respectively. CV PR-AUC results were 0.1797 ± 0.0241, 0.1910 ± 0.0271, and 0.1916 ± 0.0342 — a range of 0.012 across all three conditions. This range is smaller than the within-threshold standard deviation at every condition (0.024–0.034), indicating that threshold choice introduces less variation than fold-to-fold sampling. The 50% condition reproduces the Dataset A baseline of the missingness experiment (0.1910); both analyses used a pipeline separate from that of Table II, so absolute values differ from the 0.2083 reported there, and comparisons should be made across thresholds within this analysis. Fig. 4 visualizes CV PR-AUC stability across the three threshold conditions, indicating that the reported findings reflect properties of the manufacturing process data rather than artifacts of specific threshold selections [16].

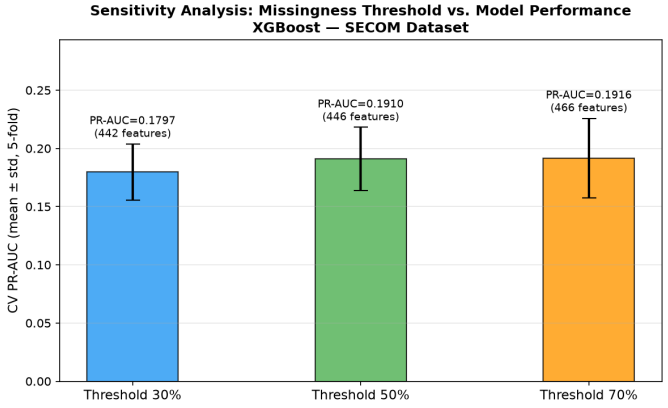


Fig. 4. CV PR-AUC across three missingness threshold conditions (30%, 50%, 70%). Error bars show ±1 standard deviation across 5 folds. The range of 0.012 across thresholds is smaller than the within-threshold standard deviation, indicating robustness to preprocessing choice.

VI. Discussion

The primary limitation of this study is sample size at the failure class level. With 104 total failures across 1,567 wafers, cross-validated test folds contain approximately 21 failures each — a count too small to distinguish model performance with statistical confidence [15]. The overlapping standard deviations in Table II reflect this irreducible uncertainty: the observed PR-AUC differences between models are within the noise of the cross-validation procedure. Future work should examine larger proprietary semiconductor datasets or apply transfer learning across multiple fab datasets to build statistically reliable yield prediction models [2].

The controlled missingness experiment produced an inconclusive result — adding 538 binary missingness indicators improved CV PR-AUC by only +0.0062 (0.1910 to 0.1972), below the pre-specified 0.01 threshold and within the noise of the cross-validation procedure. This finding has practical implications: sensor dropout in the SECOM dataset appears predominantly random rather than failure-correlated, and the median imputation strategy does not appear to discard meaningful failure signal. This null result provides a reference point for future work on this dataset, although datasets with larger failure counts may have sufficient statistical power to detect a genuine but small missingness effect [17].

The sensor masking case identified through SHAP analysis has direct implications for process monitoring system design [24]. Statistical process control systems that examine individual sensors independently cannot detect failure modes where one sensor reading suppresses failure signals from other sensors. The observation that XGBoost recovers this failure where Random Forest does not motivates development of ensemble architectures that explicitly model sensor interactions and correct for masking effects — consistent with recent directions in multivariate manufacturing intelligence research [10], [26].

VII. Conclusion

This paper presented a comparative study of logistic regression, Random Forest, and XGBoost for semiconductor yield prediction on the SECOM dataset, incorporating cross-validated evaluation, SHAP interpretability analysis, and a controlled missingness experiment. Cross-validated PR-AUC results show XGBoost marginally outperforming Random Forest (0.2083 vs. 0.1956), though differences are within noise bounds given the dataset's 104 failures. Per-wafer analysis shows that XGBoost recovers 19 of 21 test failures, including a sensor masking case missed by Random Forest, at the cost of substantially more false alarms; the missingness hypothesis was not supported at this sample size. These findings motivate two directions for future work: a stacking ensemble combining Random Forest's lower false-alarm rate with XGBoost's sequential error correction to reduce false alarms while maintaining high recall, and replication of this analysis on larger proprietary semiconductor datasets where statistical power is sufficient to detect small but operationally meaningful performance differences.

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